

Data Acquisition Systems

Test Anything, Anywhere

Data acquisition systems have evolved to provide more accurate, fast and repetitive data. With these, you can acquire any signal from anywhere and produce reliable results



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Data acquisition is the most important task for any system or application that involves signals. Acquiring the accurate data and analysing it correctly is critical for applications ranging from health-care, automation and control to space exploration. You need to acquire data to test a system, make useful analysis and give feedback signal to the system. It is the first and most important step in any project.

All the applications have their own requirements of speed of acquiring, transferring and representing the data. Keeping in sync, there has been a sub-

stantial advancement in data acquisition systems (DAQs). "Use of wireless technology, image processing algorithms, and faster and higher-resolution analogue-to-digital converters (ADCs) has increased. New DAQ components have made the systems smaller, faster, cheaper and able to pack in more applications," shares Shirish Patwardhan, CTO, KPIT Cummins.

"Intelligent DAQs with inbuilt field-programmable gate arrays (FPGAs), developments in bus technologies and virtual instrumentation are some of the latest and fast-moving developments in data acquisition systems," informs Pradeep Nair, business development manager, National Instruments.

Arif Fahim from AD Instruments explains, "Computer-based data systems reduce a lot of effort put in manual calculations and help in producing error-free analysis."

Components of data acquisition systems like sensors, transducers, ADCs, interfaces and application software have undergone a revolutionary change, making DAQ systems faster and more reliable.

Sensors go wireless

Sensors are the first point of contact between the system under observation and the DAQ system. Transducers sense physical phenomena like temperature, pressure and strain and convert these into useful electric signals that are actually measured by the DAQ system. These signals undergo signal conditioning depending on the quality of the signal acquired.

As wireless sensors are becoming popular, Nair cites the example of monitoring the flooding Yamuna. He explains, "You can place the sensors along the bank to measure the temperature and water level and push the data to a central server without a wired connection. Earlier, these projects were not touched due to infeasibility and cost involved in laying long cables along the river."

Fahim feels that wireless sensor technology is very useful for data acquisition in life science research.

"Sensors available today are much more accurate and sensitive and do a fantastic job. There are smart sensors with built-in intelligence, radio frequency identification (RFID) based sensors and micro-electromechanical